

On the Divisibility of Linear Codes and Arcs

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1. Introduction

Definition. A linear $[n, k]_q$ code is called **divisible** with **divisor** Δ if the weight of every codeword is divisible by Δ .

Definition. An (n, w) -arc $\mathcal{K}$ in $\text{PG}(k-1, q)$ is called divisible with divisor Δ if the weight of every codeword is divisible by Δ .

H. N. Ward: If $(\Delta, q) = 1$, then a code of full length (no coordinate identically zero) divisible by Δ is a Δ -fold replicated code, i.e. a code obtained from a shorter code by repeating each coordinate Δ times.

Denote by $n_q(k, d)$ the minimal length of a linear code of fixed dimension k and fixed minimum distance d over a field with q elements. A lower bound on $n_q(k, d)$ is given by:

$$n_q(k, d) \geq g_q(k, d) := \sum_{i=0}^{k-1} \left\lceil \frac{d}{q^i} \right\rceil. \quad (1)$$

This bound was proved originally by Griesmer for binary codes and generalized for q -ary codes by Solomon and Stiffler.

An $[n, k, d]_q$ -code with $n = g_q(k, d)$ is called a **Griesmer code**. The Griesmer codes exhibit strong divisibility properties.

Let

$$d = sq^{k-1} - \lambda_{k-2}q^{k-2} - \dots - \lambda_1q - \lambda_0, \quad 0 \leq \lambda_i \leq q-1, \quad (2)$$

It is easily checked that

$$n = g_q(k, d) = sv_k - \lambda_{k-2}v_{k-1} - \dots \lambda_1v_2 - \lambda_0v_1, \quad (3)$$

where $v_i = (q^i - 1)/(q - 1)$.

The arc associated with a $[n = n_q(k, d), k, d]_q$ code has parameters (n, w) with

$$w = sv_{k-1} - \lambda_{k-2}v_{k-2} - \dots \lambda_1v_1. \quad (4)$$

2. Brief history

Dodunekov and Manev (1984): for a binary Griesmer code the highest power of the minimum distance code is also divisor of the code itself.

This result was extended to linear codes over arbitrary prime fields.

Theorem 1. (H. N. Ward) Let C be a Griesmer code over $\mathbb{F}_p$, p a prime. Then if p^e divides the minimum weight of C , p^e is a divisor of the code.

Theorem 2. (H. N. Ward) Let C be a $[g_q(k, d), k, d]_q$ Griesmer code, and suppose that q divides d . Then C is divisible by p , the prime dividing q .

Theorem 3. (H. N. Ward) Let C be a $[g_4(k, d), k, d]_4$ -code for which 2^e divides d with $e \geq 2$. Then 2^{e-1} is a divisor of C .

Conjecture. (H. N. Ward) Let C be a $[g_q(k, d), k, d]_q$ Griesmer code where q is a power of the prime p . Suppose that p^e divides d with $p^e > q$. Then C is divisible by p^{e+1}/d .

Questions.

- (1) How does Ward's theorem generalize for codes over **non-prime** fields (for arcs in geometries of non-prime order)?
- (2) How does Ward's result generalize for **non-Griesmer** codes (arcs associated with non-Griesmer codes)?
- (3) How does Ward's theorem generalize if we assume that the multiplicities of the hyperplanes (the weights of the codewords) give **different** residues modulo the divisor.

3. A Divisibility Theorem

Let $\mathcal{K}$ be an arc of cardinality n in $\text{PG}(k-1, q)$, and let $e \geq 1$ be an integer.

Definition. A subspace S of dimension $k-3$ (co-dimension 2) is said to have the property (D) if every hyperplane H through S satisfies $\mathcal{K}(H) \equiv n \pmod{q^e}$.

Note: We do not require $\mathcal{K}$ to be a Griesmer arc.

For Griesmer arcs with divisor q^e all subspaces of maximal multiplicity of codimension 2 have property (D) .

Theorem 4. Let $\mathcal{K}$ be an (n, w) -arc in $\text{PG}(k - 1, q)$, and let $e \geq 1$ be an integer. If there exists a subspace δ of codimension 3 with the properties

- (1) $\mathcal{K}(\delta) \equiv n \pmod{q^{e-2}}$ if $e \geq 2$;
- (2) for every subspace S of codimension 2 through δ it holds $\mathcal{K}(S) \equiv n \pmod{q^{e-1}}$;
- (3) there exist at least q subspaces of codimension 2 through δ with the property (D).

then $\mathcal{K}(H) \equiv n \pmod{pq^{e-1}}$ for every hyperplane H containing δ .

Lemma. For a Griesmer (n, w) -arc in $\text{PG}(k - 1, q)$, every hyperplane with a point of maximal multiplicity contains a subspace of codimension 3 that satisfies properties (1)–(3).

Theorem 5. Let C be a $[g_q(k, d), k, d]_q$, $q = p^h$, be a Griesmer code and let d be given by (2) where $s = 1$, and suppose that q^e , $e \geq 2$, divides d . Then C is divisible by $p^2 q^{e-2}$.

For $q = p$, this result implies the classical Ward theorem in the special case of projective codes.

4. Quasidivisibility

Definition. We call an (n, w) -arc $\mathcal{K}$ in $\text{PG}(k-1, q)$ t -quasidivisible with divisor Δ if $\mathcal{K}(H) \equiv n, n+1, \dots, n+t \pmod{\Delta}$ for every hyperplane H .

Definition. An $[n, k, d]_q$ -code C is called t -quasidivisible if $\text{wt}(c) \equiv -t, -(t-1), \dots, 0 \pmod{\Delta}$ for every codeword $c \in C$.

As before, t -quasidivisible arcs are associated with t -quasidivisible codes.

For $t = 1$, we say that $\mathcal{K}$ (resp. C) is just quasidivisible.

By Hill-Lizak's extension lemma every quasidivisible arc (resp. quasidivisible code) with divisor q is extendable to a divisible arc (resp. divisible code) with the same divisor.

5. Divisibility and quasidivisibility results

Let S be a subspace of codimension 2 in $\text{PG}(k-1, q)$ and let $H_0, \dots, H_q$ be the hyperplanes through S .

Definition. The subspace S is said to have the property (QD) if :

- $\mathcal{K}(H_i) \equiv n + 1 \pmod{q}$ for q of the hyperplanes through S ,
- $\mathcal{K}(H_i) \equiv n \pmod{q}$ for exactly one hyperplane H_i .

Theorem 6. Let $\mathcal{K}$ be a (n, w) -arc in $\text{PG}(k-1, p)$, p -prime, with $w \equiv n+1 \pmod{p}$. Assume δ is a subspace of codimension 3 such that there exist at least $2p-1$ subspaces of codimension 2 through δ having the property (QD) with $q = p$. Then for every hyperplane H through δ it holds

$$\mathcal{K}(H) \equiv n \text{ or } n+1 \pmod{p}.$$

Theorem 7. Let $\mathcal{K}$ be a Griesmer (n, w) -arc in $\text{PG}(k-1, p)$, p -prime, with $w \equiv n+1 \pmod{p}$. Let

$$d = n - w = sp^{k-1} - \lambda_{k-2}p^{k-2} - \dots - \lambda_1p - 1,$$

with $\lambda_i \leq p-2$ for $i = 1, \dots, k-2$. Then $\mathcal{K}$ is 1-quasidivisible and hence extendible to a divisible Griesmer arc with divisor p .

5. An Example

Consider a blocking set $\mathcal{B}$ in $\text{PG}(2, q)$ with parameters $(x(p+1)+1, x)$, $x \leq p-1$.

Denote by Λ_i the number of points of multiplicity i . By Theorem 7 applied for the arc $\mathcal{K} = 2 - \mathcal{B}$ if $\Lambda_0 \geq 2p-1$, $\mathcal{B}$ is reducible to a canonical $(x(p+1), x)$ -blocking set. This is always the case if $x \leq p-2$.

For $x = p-1$ irreducible blocking sets are possible.

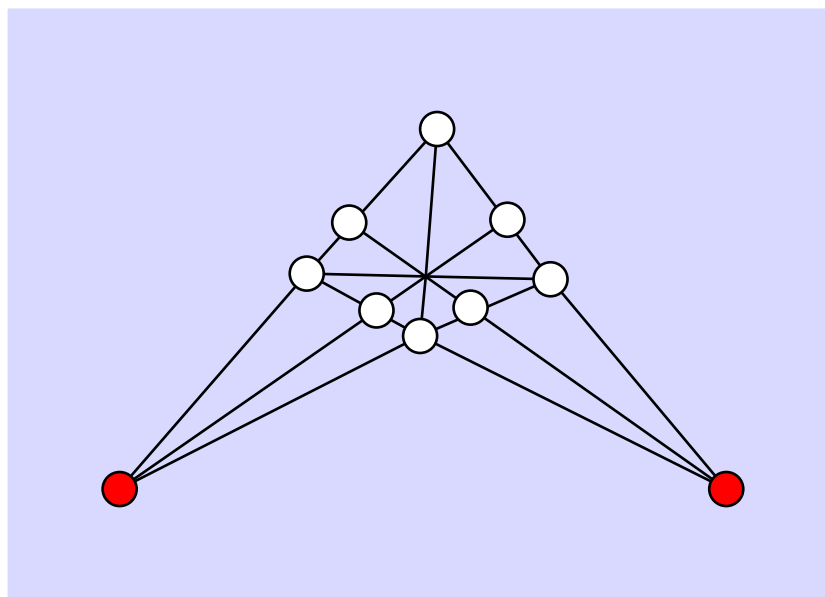
For an irreducible $(p^2, p-1)$ -blocking set $\mathcal{B}$ the undecided cases are $p+1 \leq \Lambda_0 \leq 2p-2$.

If $\Lambda_0 = p+1$ then $\mathcal{B}$ is the complement of an oval.

The classification for $p + 2 \leq \Lambda_0 \leq 2p - 2$ is unknown.

There exist $(p^2, p - 1)$ -blocking sets for which the number of 0-points is in this interval. For instance, there exists a $(25, 4)$ -blocking set in $\text{PG}(2, 5)$ with $\Lambda_0 = 8$ ($\Lambda_2 = 2$).

$(25, 4)$ -blocking set in $\text{PG}(2, 5)$ with $\Lambda_0 = 8, \Lambda_2 = 2$



$(p^2, p - 1)$ -blocking set in $\text{PG}(2, p)$ with $\Lambda_0 = 2p - 2, \Lambda_{p-2} = 1$

